

Real Estate Agency Portal

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Introduction

This project is a real estate web app where users can browse properties, contact agents, and do things like buying or renting a place. The main goal was to build a system that connects a website with a database so everything works together smoothly and stores data properly.

Instead of doing everything manually, this system makes it easier to manage properties and user interactions in one place. It simulates how a real real estate website works, where users can search, view details, and take actions on properties.

The app has different types of users like agents, buyers, and renters, and each one has different permissions. This makes the system more realistic and organized.

System Features

This system has a bunch of features that make it feel like a real application and not just a basic project.

User System

Users can sign up and log in to the system. During registration, they choose a role such as agent, buyer, or renter. Each role controls what the user is allowed to do.

Passwords are not stored directly. Instead, they are hashed using `password_hash`, which makes the system more secure. When logging in, the system uses `password_verify` to check the password.

Role Based Access

Not all users can do the same actions, which makes the system more organized.

- Agents can add new properties and delete properties they manage and edit them
- Buyers and renters can browse properties, save favorites, send inquiries, and perform transactions

This separation makes sure users only see and do what they are supposed to.

Property Management

Agents can add properties by entering details like:

- Title
- Property type
- Address
- City
- Price
- Images

Once added, the property appears in the listings page. Each property can also have two images, the interior and exterior image, which improves how it looks to users.

Users can click on any property to see full details, including all the information and images.

Search Feature

Users can search for properties using filters such as:

- City
- Minimum price
- Maximum price

This feature helps users find properties faster instead of scrolling through everything. It makes the system more user friendly.

Favorites System

Users can save property they are interested in. These saved properties appear in a separate Favorites page. This is useful because users don't have to search again if they want to revisit a property later.

Inquiry System

Users can send messages about a property. For example, they can ask questions or request more details.

These inquiries are saved in the database and linked to both the user and the property. This allows agents to later see who is interested in their properties.

Transactions: Buy and Rent

Users can buy or rent a property directly from the system.

When this happens:

- A transaction is saved in the database
- The property status changes to sold or rented
- The property is no longer available for other users

This makes the system behave more like a real world application.

Database Design

The system uses a relational database with multiple tables that are connected together.

Main Tables

- **Users** which stores user information like username, password, role, and contact info
- **Properties** which stores all property details
- **Transactions** which stores records of buying and renting
- **Inquiries** which stores messages sent by users
- **Favorites** which stores saved properties

Each table is connected using foreign keys. This helps keep the data consistent and prevents errors.

Stored Procedure and Trigger

To make the system more advanced and efficient, I used database features like stored procedures and triggers.

Stored Procedure

I created a stored procedure called **ProcessTransaction**.

This procedure:

- Inserts a transaction into the database
- Updates the property status automatically

This makes the system cleaner because the logic is handled inside the database instead of repeating code in PHP.

Trigger and View

I also used a trigger that runs automatically after a transaction is inserted. The trigger updates the property status based on whether the property was bought or rented. This helps keep the database accurate without needing extra manual updates.

In addition, I used a view instead of writing joins directly in my queries. The view combines data from multiple tables, like properties and agents, so it is easier to display property listings in a clean and simple way.

How the System Works

The system follows a really simple flow:

1. A user registers and logs in
2. The user browses available properties
3. The user can:
 - Save properties
 - Send inquiries
 - Buy or rent a property
4. When a transaction happens:
 - The transaction is stored
 - The property status is updated
 - The property is no longer available

5. Agents can manage properties by adding, deleting, or editing them

Problems and Fixes

While building this project, I ran into a few issues:

- **Stored procedure not working at first**

I realized it was not created in the correct database, so I fixed that and it started working.

- **Error when deleting properties**

This happened because of foreign key constraints. I solved it by deleting related records like transactions, inquiries and favorites before deleting the property.

- **Buy and Rent buttons still showing after purchase**

I fixed this by checking the property status before showing the buttons.

- **Images not deleting automatically**

I handled this manually by just removing the files from the uploads folder.

These issues helped me understand how databases and backend logic work together and how to debug problems step by step. But overall this project turned out really really solid. It connects PHP and MySQL to build a full working real estate system with features like login, property listings, and transactions. It helped me understand how backend systems work, how databases connect to web applications, and how to solve real problems when things don't work the first time.

Real Estate Agency Portal

[Home](#) [Properties](#) [Login](#) [Register](#)

Welcome

This is the starter version of the Real Estate Agency Portal for Spring 2026.

Use this project to practice PHP, MySQL, sessions, authentication, and CRUD operations.

Project Roles

- Agent: add and manage listings
- Buyer: browse properties and submit inquiries
- Renter: browse properties and submit inquiries

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Login

Username

Password

Login

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Register

Username

Contact Info

Password

User Type

Select role

Select role

Agent

Buyer

Renter

Register

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[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

City

Min Price

Max Price

Search

Property Listings



Modern 2026 Condo

Type: Condo

Address: 789 Pine Rd

City: Chicago

Price: \$3000000.00

Status: rented

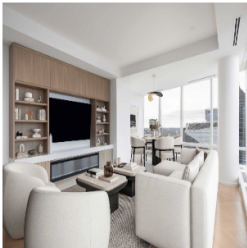
Agent: agent_Nasir

[View Details](#)

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[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

Property Details



Modern 2026 Condo

Type: Condo
Address: 789 Pine Rd
City: Chicago
Price: \$3000000.00
Status: rented
Agent: agent_Nasir



Luxury Apartment

Type: Apartment

Address: 250 Bedford Park Boulevard

City: Bronx

Price: \$5000.00

Status: available

Agent: agent_Nasir

[Submit Inquiry](#) [Save to Favorites](#)

Buy Property

Rent Property

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[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

Dashboard

Welcome: buyer_Naz

Role: buyer

Common Actions

[Browse Properties](#)

[My Favorites](#)

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My Favorites



Modern 2026 Condo

City: Chicago

Price: \$3000000.00

Agent: agent_Nasir

[View Details](#)

[Remove](#)

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Login

Invalid username or password.

Username

Password



Login

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[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

Dashboard

Welcome: agent_Nasir

Role: agent

Agent Actions

[Add Property](#)

[View Inquiries](#)

Common Actions

[Browse Properties](#)

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[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

Add Property

Title

Property Type

Address

City

Price

Status

available

▼

Exterior Image

Choose File

No file chosen

Interior Image

Choose File

No file chosen

Add Property

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My Property Inquiries

Property: Luxury Apartment User: buyer_Naz Message: Is this still available? Date: 2026-04-11 23:30:39
Property: Luxury Apartment User: buyer_Naz Message: How much for one week stay? Date: 2026-04-29 14:20:59
Property: Family House User: renter_Ahmed Message: Can I schedule a visit? Date: 2026-04-11 23:30:39



Family House

Type: House
Address: 456 Oak Ave
City: Los Angeles
Price: \$750000.00
Status: sold
Agent: agent_Nasir
[View Details](#) [Edit](#) [Delete](#)

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[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

Property Details



Family House

Type: House

Address: 456 Oak Ave

City: Los Angeles

Price: \$750000.00

Status: sold

Agent: agent_Nasir

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[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

Edit Property

Modern 2026 Condo	Condo	789 Pine Rd
Chicago	3000000.00	available Update

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[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

Dashboard

Welcome: agent_abu

Role: agent

Agent Actions

[Add Property](#)

[View Inquiries](#)

Common Actions

[Browse Properties](#)

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[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

My Property Inquiries

No inquiries yet.
